

Define Problem Statement

Project Title: Rising Waters

Introduction

Floods are one of the most dangerous natural disasters, causing loss of lives, damage to property, agriculture, and infrastructure. Heavy rainfall and changing weather conditions increase the risk of flooding. Early prediction is essential to reduce damage and improve disaster preparedness.

Problem Statement

Traditional flood prediction methods rely on manual observation and historical data, which may not always provide timely or accurate results. This makes it difficult for authorities and the public to take preventive measures. Therefore, an intelligent system is required to analyze weather data and predict flood risk efficiently.

Proposed Solution

Rising Waters is a Machine Learning-based Flood Prediction System that predicts flood risk using weather parameters such as temperature, humidity, cloud cover, and rainfall data. The system analyzes these inputs using a trained ML model and displays the prediction through a simple web application.

Objectives

- Develop an intelligent flood prediction system.
- Provide early flood warnings.
- Improve prediction accuracy using Machine Learning.
- Support disaster management and public safety.
- Reduce flood-related losses.

Expected Outcome

The system provides quick and reliable flood predictions, helping users and authorities take preventive actions before flood situations become severe.

Conclusion

Rising Waters offers a smart and efficient solution for flood prediction by combining Machine Learning with a web application. It supports early warning, improves disaster preparedness, and helps reduce the impact of floods.